

Mini Project: Training Institute Management System

Project Title

Training Institute Management System using Java & Advanced Java

Objective

Develop a dynamic application to manage:

- Student registrations
- Course allocation
- Faculty management
- Fee records
- Attendance
- Login authentication

The project should demonstrate Core Java, JDBC, Servlets/JSP, and Database concepts.

Technologies Required

Layer	Technology
Frontend	HTML, CSS, JSP
Backend	Java, Servlets
Database	MySQL or Oracle Database
Connectivity	JDBC
Server	Apache Tomcat

Functional Requirements

Module 1: User Authentication

Features:

- Admin Login
- Student Login
- Faculty Login
- Session handling
- Logout functionality

Skills Checked:

- Servlets

- Sessions
- JDBC validation

Module 2: Student Management

Features:

- Add Student
- Update Student
- Delete Student
- Search Student
- View All Students

Fields:

- Student ID
- Name
- Email
- Course
- Contact Number

Skills Checked:

- CRUD operations
- PreparedStatement
- Exception handling

Module 3: Course Management

Features:

- Add Course
- Assign Faculty
- Course Duration
- Fees Structure

Skills Checked:

- OOP concepts
- Database relationships

Module 4: Attendance Management

Features:

- Mark Attendance
- View Attendance Report
- Attendance Percentage Calculation

Skills Checked:

- Loops
- JDBC
- Business logic

Module 5: Fee Management

Features:

- Record Fee Payment
- Pending Fees Report
- Generate Receipt

Skills Checked:

- Transactions
- SQL Queries

Database Tables

STUDENTS

Column Name	Data Type
student_id	NUMBER/INT
student_name	VARCHAR
email	VARCHAR
course	VARCHAR
phone	VARCHAR

COURSES

Column Name	Data Type
course_id	NUMBER/INT
course_name	VARCHAR
duration	VARCHAR
fees	NUMBER

ATTENDANCE

Column Name	Data Type
attendance_id	NUMBER/INT
student_id	NUMBER/INT

Column Name	Data Type
date	DATE
status	VARCHAR

FEES

Column Name	Data Type
payment_id	NUMBER/INT
student_id	NUMBER/INT
amount_paid	NUMBER
payment_date	DATE

Advanced Java Concepts Required

Concept	Usage
JDBC	Database connectivity
Servlets	Request handling
JSP	Dynamic pages
Session Management	Login sessions
MVC Architecture	Project structure
Exception Handling	Error management
Connection Pooling	Optional advanced feature
DAO Pattern	Clean coding structure

Additional Tasks

1. Export Student Report to PDF
2. Email Notification after Registration
3. Search using AJAX
4. Dashboard with Charts
5. Upload Student Photo

Expected Outcome

The application should:

- Run on Apache Tomcat
- Connect successfully with the database
- Perform all CRUD operations
- Handle user sessions properly
- Follow clean coding standards

Skills Evaluated

- Core Java
- JDBC
- Servlets & JSP
- SQL
- OOP
- Exception Handling
- Mini ERP Design Concepts

